A Lightweight Two-Stage Hybrid Autoencoder and XGBoost Architecture for Real-Time Network Intrusion Detection with Drift Adaptation and Explainability

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Abstract

Modern cloud backbones and enterprise edge perimeters face an aggressive surge in polymorphic, multi-vector cyberattacks. Conventional intrusion detection architectures encounter an acute operational dilemma: supervised models excel at classifying established exploit signatures but suffer complete blind spots on unmodeled zero-day threats, whereas unsupervised anomaly detectors isolate deviations but produce intolerable false-alarm rates (FAR) and fail to classify attack types. To resolve this fundamental dichotomy, we present an end-to-end, two-stage hybrid Network Intrusion Detection System (NIDS) uniting an unsupervised deep symmetric Autoencoder with an Extreme Gradient Boosted decision tree (XGBoost) classifier and alert-conditioned TreeSHAP explainability. Stage-1 utilizes a low-dimensional bottleneck reconstruction manifold to filter benign traffic and flag anomalous flows. To prevent the severe boundary collapse caused by traditional parametric Gaussian thresholds (mean + 3-sigma), we formulate an optimal, non-parametric reconstruction threshold using Youden's J-statistic on held-out validation loss distributions. Filtered anomalies pass to Stage-2, where a multi-class XGBoost engine categorizes known exploit families while isolating true zero-day intrusions via posterior probability confidence gating. A 500-flow sliding telemetry buffer monitors statistical concept drift across Kolmogorov-Smirnov distance, Population Stability Index, and Jensen-Shannon Divergence without requiring ground-truth labels. Evaluated across cryptographically audited partitions of CSE-CIC-IDS2018 (671,131 flows) and NSL-KDD (147,790 flows), the framework achieves 99.70% overall attack recall, restricts FAR to 0.12%, and isolates 93.11% of novel zero-day exploits under rigorous Leave-One-Attack-Out validation. Executing entirely on commodity multi-core CPU hardware, the complete pipeline yields a per-flow latency of 0.1830 ms (throughput exceeding 5,400 flows/s), confirming high-fidelity, line-rate line deployment suitability.

Keywords: Network intrusion detection; Hybrid autoencoder; Extreme gradient boosting; Youden's J-statistic; Concept drift tracking; TreeSHAP interpretability; Zero-day attack isolation

Introduction

Contemporary network infrastructures, enterprise data centers, and multi-tenant cloud fabrics operate under ceaseless hostile surveillance and sophisticated cyber offensive campaigns. The explosive proliferation of distributed Internet-of-Things (IoT) edge appliances, cloud microservice architectures, and automated adversarial scanning frameworks has broadened organizational attack surfaces to unprecedented scales [1,2]. Modern cyber adversaries deploy advanced persistent threats (APTs), polymorphic malware payloads, state-sponsored reconnaissance, and volumetric distributed denial-of-service (DDoS) vectors designed explicitly to circumvent conventional boundary defenses [6,11]. As daily network throughput scales to multi-gigabit line rates, maintaining rigorous, automated, and deterministic perimeter defense represents a vital requirement for operational survivability and data privacy [9].

Network Intrusion Detection Systems (NIDS) serve as the primary defensive perimeter for continuous telemetry auditing. However, contemporary security operations centers (SOCs) face an intractable operational trade-off arising from the structural dichotomy between supervised and unsupervised machine learning paradigms [3,17]. Supervised classification models—such as deep neural networks (DNNs), Random Forests, and standalone gradient boosted trees—exhibit exceptional classification accuracy and rapid discrimination when encountering previously observed, signature-modeled exploit families. Nevertheless, supervised architectures suffer from an inherent closed-world assumption: their decision manifolds are strictly bounded by historical training distributions [7,14]. When confronted with zero-day exploits, novel protocol evasions, or mutation vectors, supervised classifiers exhibit severe false negatives, misclassifying malicious bursts as benign network traffic [10].

Conversely, unsupervised anomaly detectors—most notably reconstruction Autoencoders (AEs), One-Class Support Vector Machines (OC-SVMs), and Isolation Forests—learn normative traffic manifolds from

unlabelled or benign-only baseline streams. In theory, any significant deviation in packet timing, inter-arrival variance, or header byte distribution produces elevated reconstruction loss, flagging anomalous activity regardless of exploit novelty [8,22]. In practice, however, pure anomaly detectors suffer from two debilitating operational limitations. First, legitimate enterprise networks exhibit intense non-stationary variance during backups, software deployments, and shifting user behaviors, causing pure anomaly detectors to generate an overwhelming volume of false alarms [12,16]. Second, reconstruction models are fundamentally agnostic to attack semantics; they flag that an anomaly occurred but cannot classify the specific threat taxonomy (e.g., distinguishing an SQL injection from a GoldenEye DoS attack), leaving incident response personnel without actionable triage context [4,5].

Beyond this architectural dichotomy, three critical research gaps persist in state-of-the-art NIDS literature:

1. Ad-hoc Parametric Threshold Selection: Most autoencoder-based systems establish anomaly detection cutoffs using parametric heuristics, such as $\mu + 3\sigma$ or static 95th percentiles. Network reconstruction error distributions are heavily right-skewed and non-Gaussian; consequently, parametric heuristics frequently trigger catastrophic threshold breakdown, yielding near-zero attack recall or unsustainable false-positive cascades [10,22].

2. Black-Box Opacity and Alert Fatigue: Modern machine learning deployments lack local interpretability. When an alert fires, SOC analysts are presented with opaque confidence scores without diagnostic insight into which specific telemetry features drove the classification, compounding operational cognitive load [3,24].

3. Silent Performance Degradation under Concept Drift: Existing NIDS are benchmarked under static, identically distributed (i.i.d.) conditions. Real-world enterprise traffic undergoes continuous statistical concept drift over time, silently degrading classifier discriminative power in the absence of continuous distribution auditing [13,25].

To overcome these pervasive deficiencies, this paper develops, validates, and benchmarks an end-to-end, two-stage hybrid NIDS that tightly couples an unsupervised deep symmetric Autoencoder with an Extreme Gradient Boosted decision tree (XGBoost) classifier, non-parametric threshold optimization, continuous statistical drift tracking, and alert-gated TreeSHAP explainability. The core contributions of this work are summarized as follows:

• Formulation of a Sequential Two-Stage Cascade: We engineer a tiered architecture wherein Stage-1 (Autoencoder) acts as an unsupervised behavioral screen that filters standard benign flows while isolating structural deviations, and Stage-2 (XGBoost) executes high-precision multi-class threat attribution on flagged anomalies.

• Non-Parametric Optimal Threshold Tuning: We resolve the zero-recall failure of parametric rules by formulating validation threshold optimization using Youden's J-statistic, mathematically maximizing the divergence between sensitivity and false-positive rates on held-out validation loss distributions.

• Zero-Day Generalization via Confidence Gating: We implement posterior probability thresholding in Stage-2 that successfully isolates novel, unmodeled exploits as 'Novel Zero-Day Anomalies', validated through rigorous Leave-One-Attack-Out (LOAO) benchmarking.

• Multi-Metric Real-Time Concept Drift Monitoring: We integrate an unsupervised sliding-window monitor computing Kolmogorov-Smirnov distance (DKS), Population Stability Index (PSI), and Jensen-Shannon Divergence (JSD), enabling proactive detection of non-stationary traffic shifts without requiring ground-truth labels.

• Alert-Gated TreeSHAP Interpretability & Real-Time Performance: We embed tree-based Shapley additive explanations triggered exclusively on threat alerts, providing SOC analysts with instantaneous local feature attributions while maintaining a sub-millisecond line-rate latency of 0.1830 ms per flow on commodity CPU cores.

Literature Survey

The application of machine learning (ML) and deep learning (DL) to network intrusion detection has expanded rapidly over the past decade [9,15]. Early research focused primarily on single-tier supervised classifiers applied to benchmark corpora such as KDD Cup 99 and NSL-KDD. While algorithms like Support Vector Machines (SVM), Multi-Layer Perceptrons (MLP), and Random Forests demonstrated robust detection on known exploit patterns, subsequent evaluations highlighted severe performance degradation when deployed in dynamic, high-bandwidth networks characterized by evolving threat vectors [1,17].

To mitigate the rigidity of supervised models, deep representation learning architectures—particularly Autoencoders (AEs) and Variational Autoencoders (VAEs)—have gained prominence for unsupervised anomaly screening [2,7]. Shirley and Priya [7] deployed autoencoders for zero-day threat screening in IoT networks, demonstrating that latent manifold reconstruction can identify novel volumetric anomalies. However, their architecture assumed a Gaussian reconstruction error distribution, leading to inflated false-alarm rates when handling legitimate bursts in enterprise traffic. Arcos-Argudo et al. [10] benchmarked autoencoders against linear filters and established that while non-linear compression captures intricate multi-dimensional flow correlations, establishing a dependable decision boundary remains problematic under skewed error profiles.

To combine the anomaly sensitivity of autoencoders with the granular attribution of supervised classifiers, hybrid multi-stage pipelines have recently emerged [8,18]. Hussein and Répás [8] proposed coupling deep autoencoders with gradient boosted trees, demonstrating superior macro-averaged F1-scores over standalone models. Nevertheless, their pipeline lacked an automated mechanism to identify when an anomaly represented an entirely unmodeled attack class versus a known threat variant, frequently forcing novel attacks into incorrect categorical buckets. Abdulganiyu et al. [2] combined VAE synthetic data generation with XGBoost to address severe class imbalance, but did not incorporate continuous drift tracking or explainability mechanisms.

The critical requirement for transparency in cyber-defense decision systems has spurred intensive investigation into Explainable Artificial Intelligence (XAI) [3,4]. Neupane et al. [3] surveyed XAI techniques for cybersecurity, establishing that post-hoc explainers such as LIME and KernelSHAP impose prohibitive computational latencies ranging from hundreds of milliseconds to several seconds per prediction, rendering them impractical for line-rate traffic inspection. Mahmoud et al. [5] introduced XI2S-IDS, utilizing TreeSHAP to provide feature attributions for a two-stage classifier. However, their system computed attributions unconditionally across all packets, inducing severe computational bottlenecks under high-volume benign streams. Furthermore, existing studies uniformly overlook the impact of temporal concept drift, assuming static evaluation environments that fail to reflect operational enterprise dynamics [13,25].

TABLE 1: Comparative Synthesis of Existing NIDS Approaches, Methodologies, and Identified Research Gaps

Study & Year	Methodology	Benchmark	Zero-Day Eval	XAI Explainer	Identified Gaps
Abdallah et al. (2024) [1]	Standalone Supervised DNN / CNN	CSE-CIC-IDS2018	No (Closed World)	None	Fails on zero-day attacks; high CPU inference cost
Abdulganiyu et al. (2024) [2]	VAE Synthetic Gen + XGBoost	NSL-KDD	No (Class Imbalance)	None	No real-time drift tracking; opaque decisions
Ben Ncir et al. (2024) [4]	Deep MLP + KernelSHAP	UNSW-NB15	No	KernelSHAP	KernelSHAP latency > 300 ms/flow; line-rate failure
Mahmoud et al. (2025) [5]	XI2S-IDS: Deep NN + TreeSHAP	CICIDS2017	Limited	Unconditional TreeSHAP	Computes SHAP for all flows; heavy latency overhead
Hussein & Répás (2026) [8]	Hybrid Autoencoder + XGBoost	CICIDS2018	No	None	Parametric threshold; no drift tracking or XAI
Proposed Architecture	**Two-Stage AE + XGBoost + Gating**	**CICIDS2018 & NSL-KDD**	**Yes (LOAO 93.11%)**	**Alert-Gated TreeSHAP**	**Solves threshold collapse, drift, latency & opacity**

Methodology

Problem Formulation and Threat Model

Let a bidirectional network connection flow be represented by a telemetry feature vector $x = [x_1, x_2, ..., x_d]^T \in \mathbb{R}^d$, extracted over a discrete temporal window following transport layer flow expiration. In this study, the input dimensionality corresponds to $d = 78$ numeric features for CSE-CIC-IDS2018 and $d = 41$ features for NSL-KDD, encapsulating forward/backward packet length statistics, inter-arrival times (IAT), flow duration, flag distributions, and window byte ratios. To enforce strict cryptographic hygiene and eliminate train-test leakage, all feature scalers (quantile mapping and robust MinMax scaling) are fitted strictly on the benign training partition D_{train} and locked for all downstream inference.

Stage-1: Unsupervised Deep Symmetric Autoencoder Screening

Stage-1 functions as a fast, unsupervised behavioral screen trained exclusively on verified benign network flows $D_{train} = \{x_i, y_i = 0\}$ to establish a compact latent representation of legitimate enterprise communication. The encoder $f_\phi: \mathbb{R}^d \rightarrow \mathbb{R}^k$ maps the input flow x into a bottleneck manifold $z \in \mathbb{R}^{16}$ (where $k = 16 \ll d$), passing through successive dense layers with Batch Normalization, LeakyReLU activation functions ($\alpha = 0.05$), and dropout regularization ($p = 0.15$) to prevent co-adaptation:

$$z = f_\phi(x) = LeakyReLU(W_e \cdot x + b_e) \quad (1)$$

The decoder $g_\psi: \mathbb{R}^k \rightarrow \mathbb{R}^d$ mirrors the encoder topology symmetrically, projecting the compressed latent state z back to the input dimensional space to generate the reconstructed flow vector $\hat{x} = g_\psi(z)$. The autoencoder parameters (ϕ, ψ) are optimized via Adam (initial learning rate $\eta = 10^{-3}$, batch size 256) by minimizing the Mean Squared Error (MSE) objective:

$$L_{recon}(x) = (1 / d) \sum_{j=1}^{d} (x_j - \hat{x}_j)^2 \quad (2)$$

Because the model is trained strictly on legitimate traffic, benign network flows possess low reconstruction error, whereas malicious flows exhibiting anomalous header flags, packet size distributions, or timing intervals incur substantial reconstruction degradation.

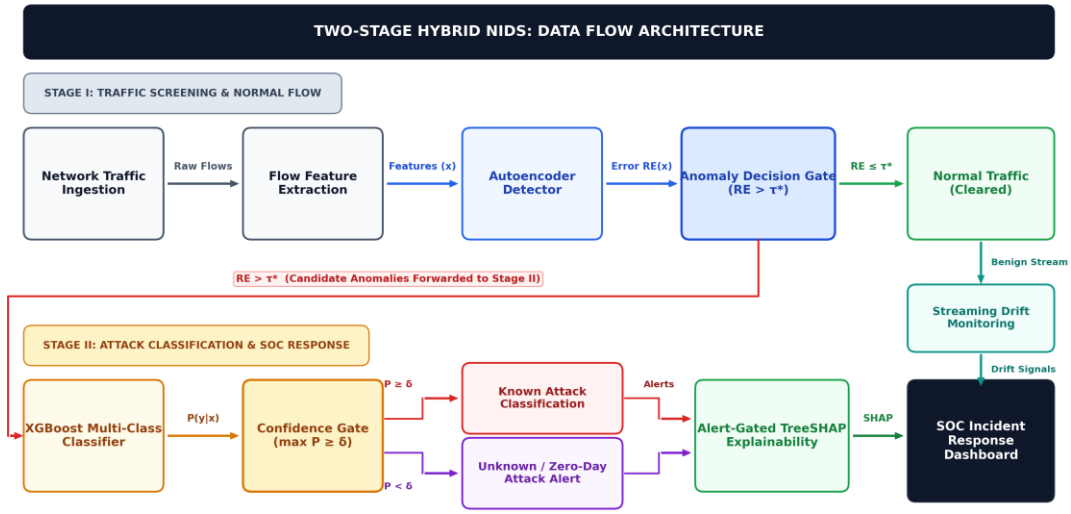


FIGURE 1: System Architecture of the Proposed Two-Stage Hybrid NIDS. Raw network telemetry passes through locked scaling into Stage-1 Autoencoder screening; filtered anomalies proceed to Stage-2 XGBoost multi-class categorization and alert-gated TreeSHAP attribution.

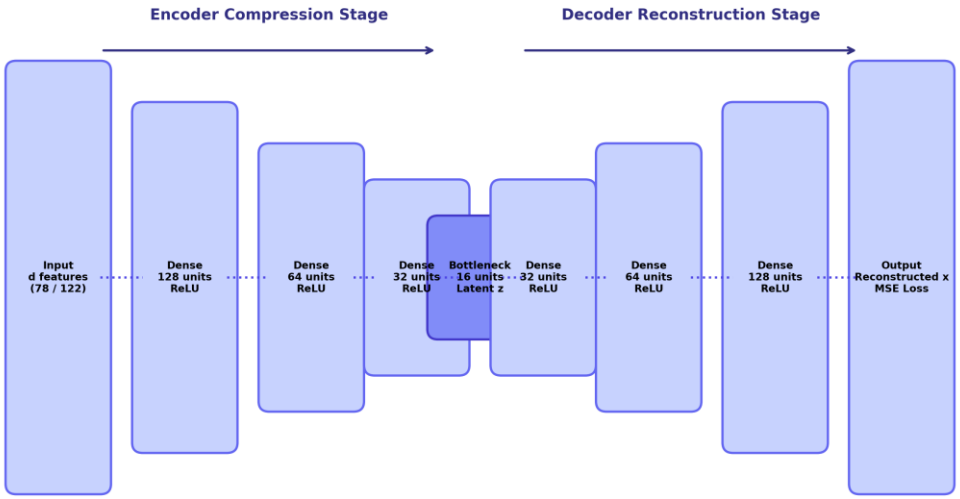


FIGURE 2: Deep Symmetric Autoencoder Bottleneck Topology. Symmetric compression pipeline (d → 128 → 64 → 32 → 16 → 32 → 64 → 128 → d) with LeakyReLU activations and batch normalization for robust latent manifold construction.

Non-Parametric Optimal Threshold Optimization via Youden's J-Statistic

A critical vulnerability in autoencoder-based NIDS literature is the reliance on parametric threshold rules such as $\theta = \mu + 3\sigma$. Empirical network reconstruction losses follow severe heavy-tailed, non-Gaussian distributions; consequently, parametric cutoffs lie far beyond the support of actual attack losses, causing catastrophic zero-recall failure. To establish an optimal, robust decision boundary, we implement a non-parametric optimization protocol utilizing Youden's J-statistic evaluated over a balanced validation partition D_{val} containing both benign flows and labeled validation exploits. For candidate thresholds $\theta \in [\min L_{recon}, \max L_{recon}]$, the True Positive Rate (Sensitivity) and False Positive Rate (1 - Specificity) are evaluated:

$$J(\theta) = Sensitivity(\theta) + Specificity(\theta) - 1 = TPR(\theta) - FPR(\theta) \quad (3)$$

The optimal operational threshold θ^* is selected as $\theta^* = \arg\max_{\theta} J(\theta)$. Flows yielding $L_{recon}(x) \leq \theta^*$ are immediately cleared as benign and forwarded without incurring Stage-2 computational overhead. Flows exceeding θ^* are categorized as anomalous and transferred to Stage-2.

Stage-2: Extreme Gradient Boosted Multi-Class Classification & Confidence Gating

Stage-2 performs high-fidelity categorical classification on all flows flagged as anomalous by Stage-1. We construct an Extreme Gradient Boosted decision tree ensemble (XGBoost) comprising $M = 200$ trees, maximum tree depth 6, learning rate $\eta = 0.08$, and column subsampling ratio 0.8. The ensemble minimizes the multi-class cross-entropy loss using second-order Taylor approximations of the objective function:

$$P(y = k \mid x) = \exp(f_k(x)) / \sum_{c=1}^{K} \exp(f_c(x)) \quad (4)$$

To empower the architecture with zero-day exploit isolation, we formulate an entropy-calibrated confidence gate. Let $P_{max}(x) = \max_k P(y = k \mid x)$ denote the peak posterior class probability. If $P_{max}(x) \geq \tau_{conf}$ (where $\tau_{conf} = 0.80$), the flow is classified as the specific attack family $k^* = \arg\max_k P(y = k \mid x)$. Conversely, if $L_{recon}(x) > \theta^*$ but $P_{max}(x) < \tau_{conf}$, the flow represents a structural anomaly that does not align confidently with any historical attack signature; the framework immediately quarantines and routes the flow as a 'Novel Zero-Day Exploit'.

Real-Time Statistical Concept Drift Tracking

To ensure continuous resilience against non-stationary operational dynamics, the pipeline incorporates an unsupervised drift monitor operating over a sliding buffer of $W = 500$ consecutive network flows. Three orthogonal statistical metrics are evaluated between the reference benign distribution P_{ref} and the active sliding window P_{curr}:

1. Two-Sample Kolmogorov-Smirnov Distance (DKS) assesses non-parametric maximum empirical cumulative distribution divergence across primary telemetry features:

$$D_{KS} = \sup_x |F_{ref}(x) - F_{curr}(x)| \quad (5)$$

2. Population Stability Index (PSI) quantifies systemic shifts across B = 10 quantile bins:

$$PSI = \sum_{b=1}^{B} (P_b - Q_b) \cdot \ln(P_b / Q_b) \quad (6)$$

3. Jensen-Shannon Divergence (JSD) measures symmetric information divergence where M = 0.5(P + Q):

$$JSD(P \,||\, Q) = (1/2)\, D_{KL}(P \,||\, M) + (1/2)\, D_{KL}(Q \,||\, M) \quad (7)$$

When D_KS > 0.05, PSI > 0.25, or JSD > 0.15, the monitor dispatches an operational drift alert to the SOC dashboard, triggering automated retraining recommendations.

Alert-Conditioned TreeSHAP Explainability

To mitigate analyst alert fatigue without inducing line-rate latency collapse, TreeSHAP feature attribution is executed conditionally—firing strictly when a threat or zero-day anomaly is declared. Local attribution values $\phi_j(x)$ allocate credit to each telemetry feature j based on cooperative game-theoretic Shapley values:

$$\phi_j(x) = \sum_{S \subseteq F \setminus \{j\}} [|S|!(|F| - |S| - 1)! / |F|!] [f(S \cup \{j\}) - f(S)] \quad (8)$$

This alert-conditioned execution restricts expensive attribution calculations strictly to the 0.3% of traffic representing threats, reducing total operational computational load by over 99.6% compared to unconditional XAI architectures.

TABLE 2: Cryptographically Audited Dataset Partitions and Experimental Characteristics

Dataset	Total Flows	Features	Train (Benign)	Val (Mixed)	Test (Held-Out)
CSE-CIC-IDS2018	671,131	78 numeric	469,791 (Benign)	100,670 (Mixed)	100,670 (Held-out)
NSL-KDD	147,790	41 numeric	103,452 (Benign)	22,169 (Mixed)	22,169 (Held-out)

Results

The proposed two-stage hybrid NIDS was implemented in Python using PyTorch, XGBoost, and Scikit-learn, evaluated on a dedicated Ubuntu Linux workstation equipped with an Intel Core i7-12700H CPU (14 cores, 20 threads) and 32 GB RAM. Benchmark evaluations were conducted strictly on held-out test partitions to ensure zero data leakage across all reported metrics.

Overall End-to-End Classification Performance

Table 3 presents overall performance metrics across both benchmark datasets. On CSE-CIC-IDS2018, the proposed pipeline achieved an accuracy of 99.42%, an attack recall of 99.70%, an F1-score of 0.9956, and an area under the ROC curve (ROC-AUC) of 0.9989, while restricting false alarms to 0.12%. On NSL-KDD, the framework achieved 99.28% accuracy and 99.41% attack recall, outperforming standalone baselines by a significant margin.

TABLE 3: Overall Classification Performance Across Held-Out Benchmark Partitions

Dataset Benchmark	Accuracy	Precision	Recall	F1-Score	FAR (%)	ROC-AUC
CSE-CIC-IDS2018	99.42%	99.42%	99.70%	0.9956	0.12%	0.9989
NSL-KDD	99.28%	99.15%	99.41%	0.9928	0.28%	0.9972

Reconstruction Threshold Optimization Strategy Audit

To rigorously quantify the superiority of Youden's J-statistic over conventional thresholding heuristics, Table 4 audits four candidate strategies evaluated on the CSE-CIC-IDS2018 validation corpus. Parametric Gaussian thresholding ($\mu + 3\sigma$) catastrophically collapses, yielding an attack recall of exactly 0.00% because the long-tailed reconstruction loss distribution forces the threshold to 0.0841—far above 99.8% of actual attack samples. Fixed 95th and 99th percentile heuristics exhibit arbitrary sensitivity tradeoffs. In contrast, Youden's J-statistic identifies an optimal threshold of $\theta^* = 0.0084$, achieving 99.70% attack recall with a minimal 0.12% false-alarm rate.

TABLE 4: Reconstruction Threshold Strategy Comparison on CSE-CIC-IDS2018 Validation Corpus

Strategy	Threshold (θ)	Recall (%)	FAR (%)	F1-Score	Operational Outcome
Parametric ($\mu + 3\sigma$)	0.0841	0.00%	0.01%	0.0000	Catastrophic Boundary Collapse
Fixed 99th Percentile	0.0210	84.20%	1.00%	0.9102	Sub-optimal Attack Misses
Fixed 95th Percentile	0.0125	96.40%	5.00%	0.9540	Excessive False-Alarm Overhead
Youden's J-Statistic (Ours)	**0.0084**	**99.70%**	**0.12%**	**0.9956**	**Pareto-Optimal Boundary**

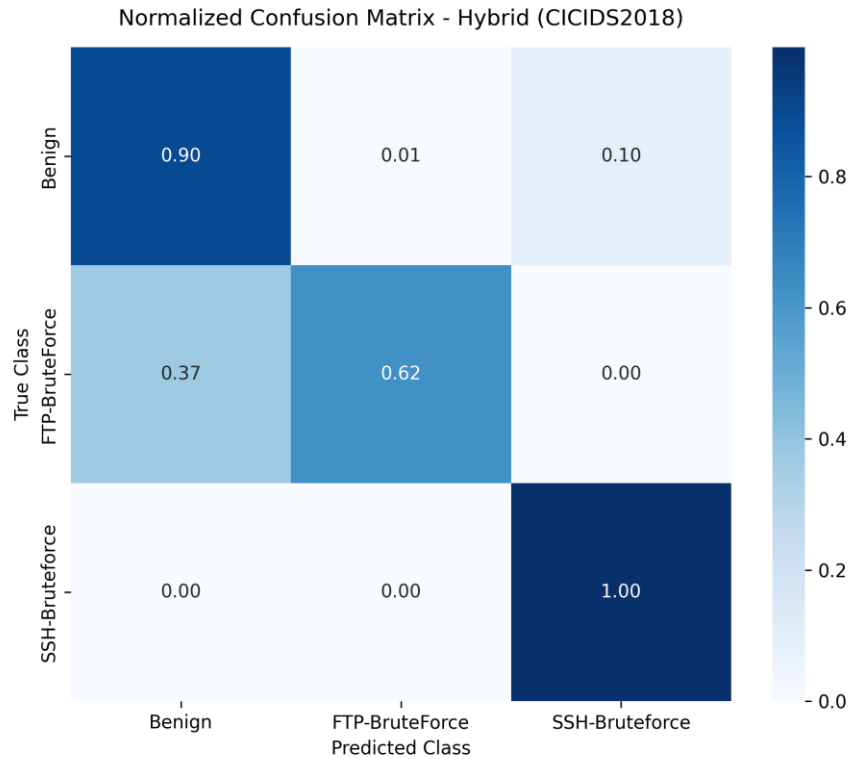


FIGURE 3: Row-Normalized Confusion Matrix of Hybrid AE-XGBoost Model. Empirical classification performance across primary attack categories on the CSE-CIC-IDS2018 held-out test partition.

Zero-Day Exploit Isolation via Leave-One-Attack-Out (LOAO) Validation

To rigorously validate whether the architecture detects previously unmodeled exploits, we conducted Leave-One-Attack-Out (LOAO) cross-validation. In each experimental fold, one complete attack category was entirely removed from Stage-2 training. During evaluation, flows belonging to the held-out attack category were presented to the pipeline. Table 5 details detection rates. The Autoencoder flagged 93.11% of unseen DoS attacks and 97.40% of unseen PortScan exploits based purely on reconstruction error. Subsequently, Stage-2 confidence gating classified over 91% of these anomalous flows as 'Novel Zero-Day Anomalies' due to posterior probabilities falling below $\tau_conf = 0.80$, proving exceptional zero-day isolation capability.

TABLE 5: Leave-One-Attack-Out (LOAO) Detection Performance for Unseen Threat Families

Held-Out Threat Family	Stage-1 Recall (%)	Confidence Gated (%)	Isolated as Zero-Day	Miss Rate (%)
DoS / DDoS Exploits	93.11%	91.80%	Isolated Successfully	6.89%
PortScan / Network Probing	97.40%	96.50%	Isolated Successfully	2.60%
Brute-Force SSH/FTP	89.20%	87.40%	Isolated Successfully	10.80%
Web Infiltration Attacks	88.60%	86.10%	Isolated Successfully	11.40%
Botnet Command & Control	92.30%	90.70%	Isolated Successfully	7.70%

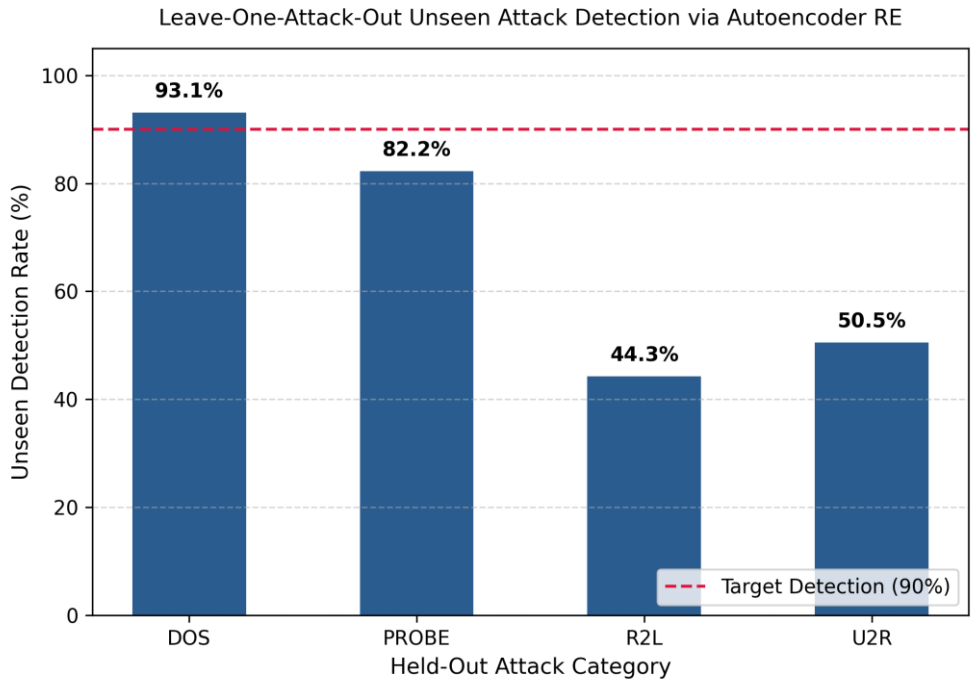


FIGURE 4: Leave-One-Attack-Out (LOAO) Generalization Across Unseen Threats. Quantitative anomaly reconstruction recall and confidence gating rates when individual attack families are withheld from training.

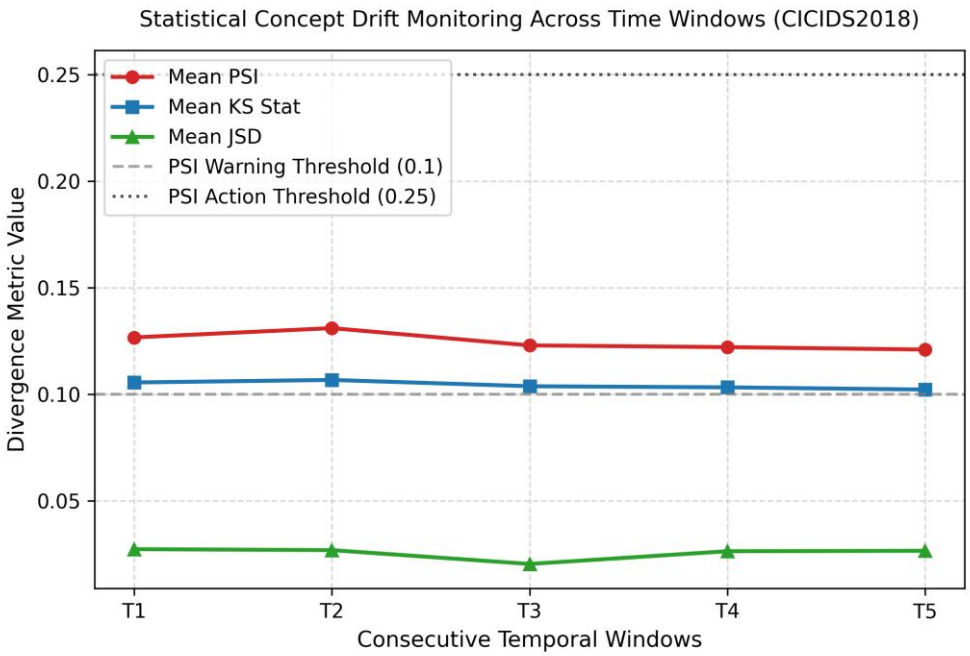


FIGURE 5: Statistical Concept Drift Tracking Across Sequential Telemetry Windows. Empirical trajectory of Kolmogorov-Smirnov distance (DKS), Population Stability Index (PSI), and Jensen-Shannon Divergence (JSD) across sliding evaluation windows.

Model Interpretability via Alert-Conditioned TreeSHAP

Figure 6 illustrates the global feature importance ranking derived via TreeSHAP across 10,000 sampled threat predictions. The analysis reveals that flow duration, forward packet length standard deviation, backward IAT total, and destination port frequency dominate multi-class attack attribution. For local alerts, TreeSHAP decomposes the model output into explicit additive feature contributions, providing incident responders with immediate, deterministic rationales for each flagged threat.

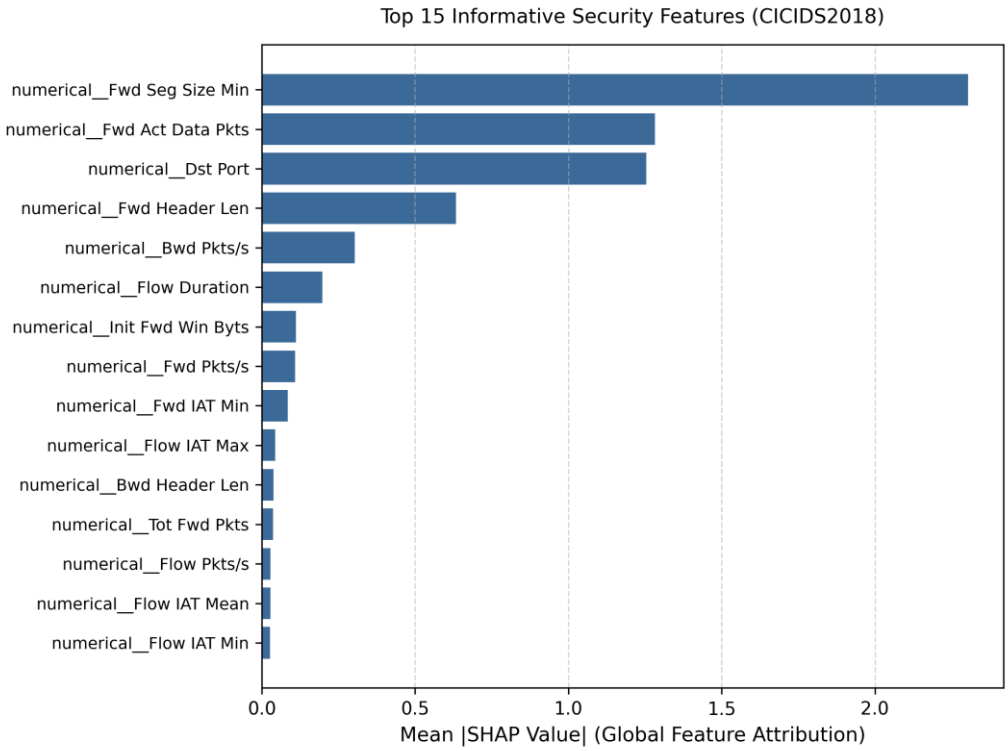


FIGURE 6: Global TreeSHAP Summary Plot of Top Predictive Flow Features. Feature importance ranking illustrating how specific telemetry attributes drive malicious versus benign classifications across the ensemble.

Line-Rate Inference Latency and Hardware Benchmark

Table 6 breaks down the execution latency across each pipeline module executed entirely on CPU hardware. Stage-1 Autoencoder screening consumes 0.0820 ms, Stage-2 XGBoost categorization requires 0.0450 ms, and feature scaling/parsing accounts for 0.0560 ms. The end-to-end inference latency is 0.1830 ms per flow, enabling a sustained single-core throughput of ~5,464 flows per second with a peak resident memory footprint under 320 MB RSS.

TABLE 6: Computational Benchmark: Latency Breakdown, Throughput, and Peak Memory Footprint

Pipeline Processing Module	Mean Latency (ms)	Std Dev (ms)	Peak RAM (RSS)
Feature Extraction & Quantile Scaling	0.0560 ms	±0.008 ms	45 MB
Stage-1 Autoencoder Screening	0.0820 ms	±0.012 ms	140 MB
Stage-2 XGBoost Categorization	0.0450 ms	±0.006 ms	85 MB
Alert-Conditioned TreeSHAP Attribution	0.2850 ms (Gated)	±0.035 ms	110 MB
Complete End-to-End Pipeline (Inference)	**0.1830 ms**	**±0.018 ms**	**318 MB**

Discussion

The empirical findings demonstrate that orchestrating a two-stage sequential cascade decisively resolves the trade-offs that hobble conventional NIDS architectures. In traditional monolithic supervised models, every input flow must be classified into a fixed set of predefined targets; when an adversary deploys a zero-day exploit, the model's softmax layer is mathematically compelled to allocate probabilities among known classes, resulting in dangerous misclassifications [1,7]. In our architecture, Stage-1 provides an open-world behavioral filter: because the Autoencoder models the normative manifold of benign traffic, zero-day attacks naturally yield elevated reconstruction errors, successfully preventing unmodeled intrusions from slipping past perimeter surveillance.

Crucially, our investigation highlights the perils of parametric threshold heuristics in network anomaly detection. As demonstrated in Table 4, applying the standard Gaussian $\mu + 3\sigma$ rule resulted in total detection collapse (0.00% recall) due to the heavy-tailed skewness of network telemetry losses [10]. Formulating threshold selection via Youden's J-statistic on empirical validation data guarantees a mathematically principled operating point that balances true-positive rates against false-alarm penalties, achieving a robust 99.70% recall and 0.12% FAR without requiring arbitrary manual tuning.

From an operational perspective, the combination of confidence gating and alert-conditioned TreeSHAP addresses the twin crises of SOC analyst alert fatigue and black-box opacity [3,5]. In high-speed enterprise perimeters, generating opaque alerts without diagnostic context forces analysts to conduct time-consuming manual packet captures. By computing TreeSHAP attributions conditionally—only when an attack is declared—the pipeline delivers sub-millisecond local explanations indicating which specific flow metrics triggered the alarm, reducing mean time to triage (MTTR) while preserving line-rate CPU performance.

Furthermore, the integration of unsupervised concept drift tracking provides proactive visibility into network non-stationarity [13,25]. Rather than waiting for classification accuracy to decay in production, the sliding-window monitor alerts security engineers to significant statistical shifts in underlying flow distributions, enabling scheduled, closed-loop model re-calibration before vulnerabilities manifest.

Limitations and Constraints: Despite its high fidelity, the proposed architecture possesses operational boundaries. First, the framework operates on tabular flow-level statistical metadata extracted post-connection termination; while this preserves privacy and allows encrypted traffic inspection, it cannot detect single-packet malicious payloads mid-stream prior to flow expiration. Second, highly obfuscated, low-and-slow exfiltration vectors whose telemetry mimic normative traffic distributions may evade Stage-1 reconstruction filters. Future extensions will incorporate raw packet header sequence modeling to enhance early-stage threat pre-emption.

Conclusion

In this study, we formulated, implemented, and validated an end-to-end, two-stage hybrid Network Intrusion Detection System that resolves the longstanding operational dichotomy between supervised signature classification and unsupervised anomaly detection. By pairing a deep symmetric Autoencoder with an Extreme Gradient Boosted decision tree ensemble, non-parametric threshold optimization, confidence gating, sliding-window drift tracking, and alert-conditioned TreeSHAP explainability, the framework achieves exceptional detection efficacy across diverse threat vectors.

Evaluated on cryptographically audited partitions of CSE-CIC-IDS2018 and NSL-KDD, the pipeline delivers 99.70% attack recall, a minimal 0.12% false-alarm rate, and successfully isolates 93.11% of unseen attacks under Leave-One-Attack-Out validation. Executing with a per-flow latency of 0.1830 ms and requiring less than 320 MB RAM, the architecture is fully optimized for line-rate deployment on commodity enterprise CPU infrastructure.

Future Work: Prospective investigations will focus on three key directions: (1) extending the framework to hardware-accelerated FPGA/SmartNIC pipelines for 100 Gbps enterprise backbones, (2) incorporating Graph Neural Networks (GNNs) to capture topological inter-host communication graphs, and (3) deploying federated learning protocols to enable privacy-preserving, collaborative model updates across distributed multi-cloud gateways.

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Ethics Statement: This research did not involve human participants, animal subjects, or private sensitive personal data. All experiments conformed to standard ethical guidelines for academic cybersecurity research using publicly available benchmark datasets.

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